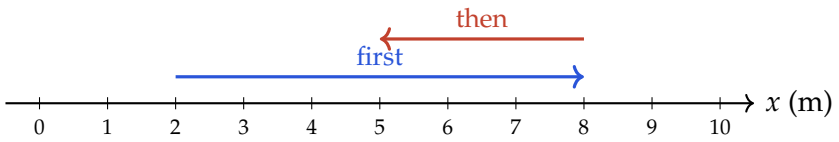
AP PHYSICS 1

Unit 1 • Lesson 1.1 • Notes (filled in)

Topic 1.1 • Scalars and Vectors in One Dimension

Notes (filled in)

Objective: tell scalars from vectors, and use signs to give the direction of a vector in one dimension.

Scalar or vector?	A scalar has a magnitude only (a number with a unit). A vector has a magnitude **and** a direction. **Scalars**: distance, speed, time, mass **Vectors**: displacement, velocity, acceleration, force
Direction in one dimension	On a line, we choose a positive direction (for example: right, east, or up). A vector pointing that way has a positive sign; the opposite way, a negative sign. The sign of a vector gives its direction, not its size: $-8\,\text{m/s}$ is faster than $+3\,\text{m/s}$.
Example: distance vs. displacement	 A cart moves from $x = 2\,\text{m}$ to $x = 8\,\text{m}$, then back to $x = 5\,\text{m}$. Distance = $6\,\text{m} + 3\,\text{m} = 9\,\text{m}$ Displacement $\Delta x = x_f - x_i = 5 - 2 = +3\,\text{m}$ Distance is the total path length (never negative). Displacement is the change in position: it only depends on the start and the end.
Adding vectors in one dimension	Add the signed values. A walker goes $+5\,\text{m}$ then $-7\,\text{m}$: total displacement = $-2\,\text{m}$, distance = $12\,\text{m}$. Two forces on a box: $+30\,\text{N}$ and $-12\,\text{N}$. Net force = $+18\,\text{N}$ (to the right).
Check	A runner does one full lap of a 400 m track and stops at the start line. Distance = $400\,\text{m}$ Displacement = $0\,\text{m}$
Summary	Scalars have a size only; vectors have a size and a direction. In one dimension the direction is the sign, once a positive direction is chosen. Distance is the path length; displacement is final position minus initial position and can be negative or zero.